Team #1234

Problem A

Your Descriptive Paper Title Here

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Abstract

We define the key quantities of the problem ($R \approx 2.5 \times 10^4\,\mathrm{km}$, $v_0 = 30\,\mathrm{km/s}$) and identify the central trade-off: if [quantity A] is too large, [failure mode 1]; if too small, [failure mode 2]. Section 1 establishes a fuel-only baseline costing 95% of initial mass. Section 2 develops the model: [governing equation] with [data-anchored fits], integrated by [method] at [step size], verified by step-halving to [n] significant figures. The admissible window is

$$\boxed{8.8296^\circ < \theta < 8.8549^\circ}$$

a margin of only 0.0253°. We therefore *advise against* the maneuver and identify [parameter] as the most promising lever to widen the window.

1 Baseline: The Cost of Doing Nothing

We first compute the cost of [baseline option]. The total energy

$$E = \tfrac{1}{2}mv_0^2 - \frac{GMm}{R} \tag{1.1}$$

is positive, hence the unmodified trajectory is hyperbolic. Circular orbit at R requires $v_f = \sqrt{GM/R}$, i.e. $\Delta v = 13.62\,\mathrm{km/s}$. From the rocket equation (derived in App. A)

$$m_f = m_0\, e^{-\Delta v/v_{ex}} \approx 121\,\mathrm{kg}, \tag{1.2}$$

a 95.15% mass cost. This anchors the comparison for Section 2.

2 Model

2.1 Governing Equations

Quadratic drag and Newtonian gravity give

$$m\ddot{\boldsymbol{r}} = -\frac{GMm}{r^3}\boldsymbol{r} \;-\; \sigma\,\frac{P(h)}{T(h)}\,|\dot{\boldsymbol{r}}|\,\dot{\boldsymbol{r}}, \qquad \sigma \equiv \frac{c_d \mu A}{2k_B}, \tag{2.1}$$

where $P(h) = \alpha e^{-h/\beta}$ and $T(h) = ah + b$ are anchored at two literature points each [**?**]. Compressing constants into σ leaves a single functional dependence P/T.

2.2 Numerical Method

Equation (**??**) is integrated by explicit Euler with $\Delta t = 10^{-3}\,\mathrm{s}$, initial condition $(x_0, y_0) = (0, R_{atm})$, entrance angle θ setting $(\dot{x}_0, \dot{y}_0)$. Inside the atmosphere both forces act; outside, gravity only; contact with the surface terminates the run. Full code in Appendix B.

2.3 Operationalizing the Criterion

A "significant" fuel saving is not quantitative enough; for a fair estimate we require $m_f \geq 0.8\,m_0$, which by inverting (**??**) bounds the exit velocity to

$$v_{min} = v_f - v_{ex}\ln(10/8), \qquad v_{max} = v_f + v_{ex}\ln(10/8). \tag{2.2}$$

3 Results

Exit speed increases monotonically with exit angle (argument in Section 3.1), so the admissible entrance angles form one interval. Bracketing by counterexample simulations:

$$8.8295^\circ < \theta_{min} < 8.8296^\circ, \qquad 8.8549^\circ < \theta_{max} < 8.8550^\circ. \tag{3.1}$$

4 Uncertainty

Halving the step to $5 \times 10^{-4}\,\mathrm{s}$ changes the exit speed in the fifth digit (16876.86 vs 16876.53 m/s); we therefore claim 4–5 significant figures, worst case (longest, near-critical trajectory) 4. Unmodeled: planetary rotation, winds up to 600 m/s [**?**], simple spacecraft shape.

5 Conclusion

The admissible window is 0.0253° wide — a 75% fuel saving (500 kg vs 2378 kg) at the price of an impractically tight corridor. We advise against the maneuver; targeting a larger orbit radius would relax the window considerably.

A Derivation of the Rocket Equation

B Simulation Code

References

[1] J. Lunine, "The atmospheres of Uranus and Neptune," *Annu. Rev. Astron. Astrophys.* **31**, 217 (1993).

[2] V. E. Suomi *et al.*, "High Winds of Neptune," *Science* **251**, 929 (1991).